IT5601
Embedded Systems and Internet of Things

Conversational Interface for Residential Automation

Done By

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Problem Statement

Smart homes are no longer just a vision for the future. They are here, and they are working. Smart devices in a home talk with each other and with the user's smartphone to deliver an automated experience that in some perspectives makes people lazy, and in others, efficient.

Are they perfect? Have they reached the desired level that we would potentially want them to? No, not really.

One of the most important issues with smart homes is that they contain different devices that are often just not compatible with each other. Some smart devices require one particular application to be installed. Others require one or more other applications to be downloaded in order to work.

Ultimately, we end up with a billion different applications for all the various smart devices we have at our home, and the worst thing is that we don't know which application to open in order to control a particular device. This makes the entire smart home experience obnoxious, making people prefer their 'regular' homes.

This is why a lot of homes in India do not contain smart accessories and devices. They are not just used to existing, old school switches and fans and lights, they actually find it better. Why waste time finding the right application, opening it, looking for the right widget or button to click, when we could just go and toggle a switch? It's faster, easier, more efficient, and more effective. Or at least, that's the case with the way smart homes are implemented right now in a majority of regions.

Existing Solutions

There have been efforts to provide a unified interface that groups all smart devices into one control platform, but the problem is that this interface now needs to be cross-compatible with all the different applications that we have installed. Instead of acting as a cure, this particular solution only acts as a roughly patched up band-aid.

The attempts to nullify the problems of the present implementation of smart homes and smart devices in general are just not working right now, because these solutions are being built on top of a base that is already crumbling, shaky at best.

Proposed Solution

The solution that we propose is one that integrates smart devices in a smart home with one single application for all devices right from the get go, that can be controlled through touch and voice.

The present implementation of the solution is dubbed the ‘Conversation Interface for Residential Automation’, or CIRA, for short. Right now, CIRA controls the lights, AC, and the door, providing various automations and integrations between these devices.

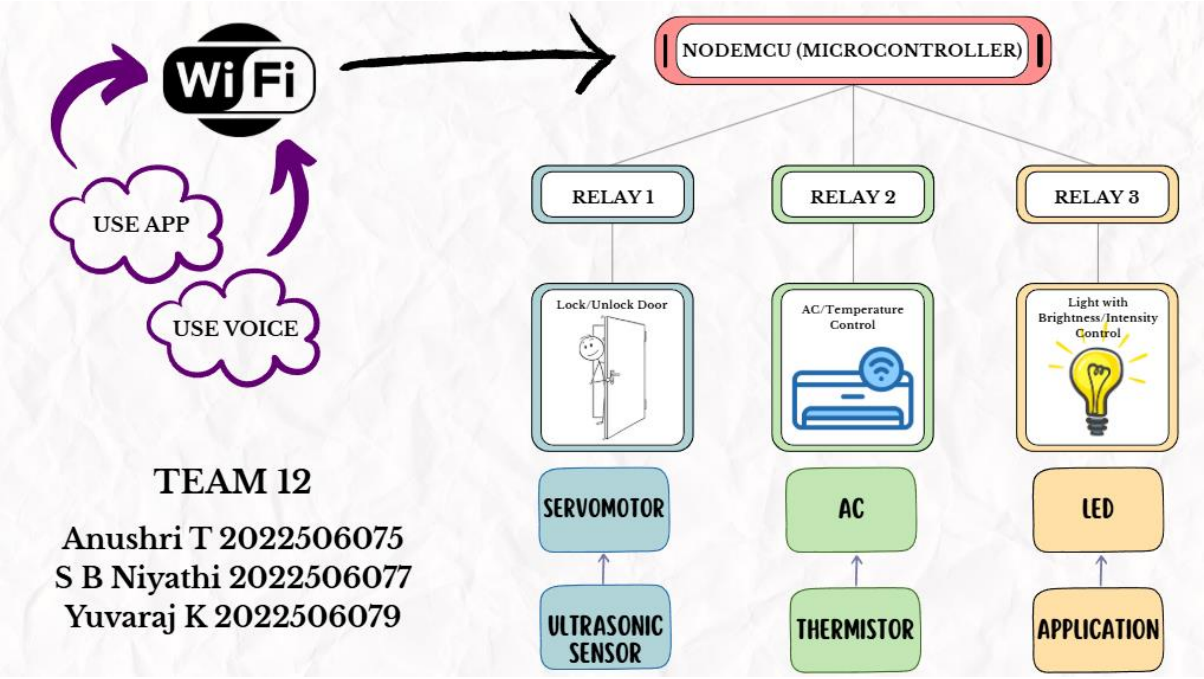
With CIRA, the lights and their intensity can be controlled. The air conditioner and its temperature can be changed. The door can be opened or closed depending on whether there is someone present outside, which also CIRA tells you. Upon the door opening, the living room lights are automatically turned on to provide better visibility for the one who has entered.

This is all implemented as a simple prototype with different sensors and actuators that elaborately mimic the actions of actual smart devices that we would be utilising in a smart home. The provided web services are asynchronous, making response time faster and enabling swift actions for the users.

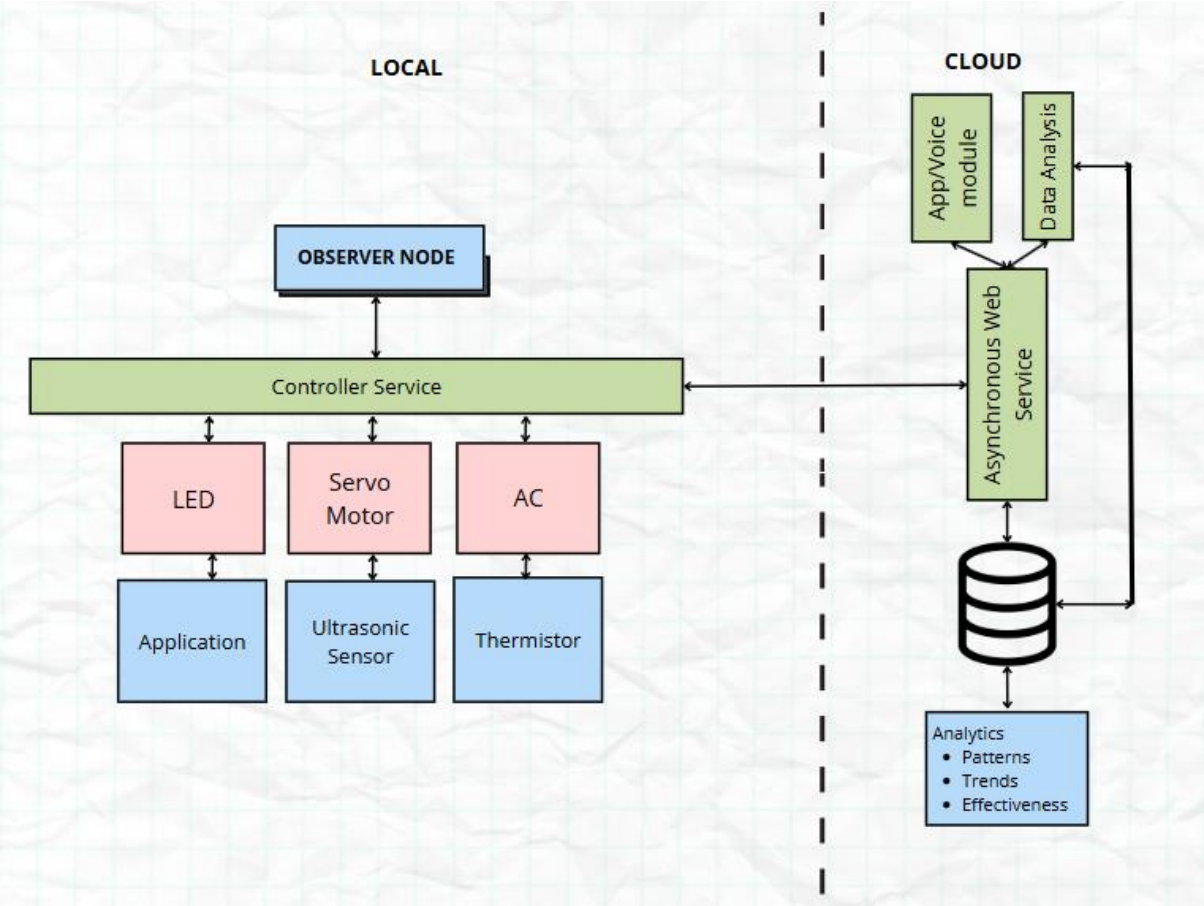
Components Required

S. No.	Component	Quantity
1	ESP8266	1
2	USB-A to Micro-USB Cable	1
3	LED	2
4	Ultrasonic Sensor	1
5	Servo Motor	1
6	Thermistor	1
7	Male-to-Male Wires	Required Amount
8	Male-to-Female Wires	Required Amount

Workflow



Level Diagram



The designed residential automation system is a level 4 IoT system consisting of multiple sensors and actuators. All the devices are controlled by the ESP8266 microcontroller, and the various kinds of information are displayed on a mobile or web interface through an asynchronous application.

Different controls allowed can be implemented through voice commands, which again flow through the web application and then to the devices and the user or observer.

Technology Stack

Frontend	HTML	Generates web application
	CSS	Provides styling
Interface Control	HTTP GET	/lightOn, /doorOpen, etc.
Action-Based UI	home(…) function	Dynamic pages based on different device states
Backend	ESP8266	Hosts web server and controls different IoT devices
	CH34x	To install device drivers that enable communication of device with ESP8266
	Arduino IDE	To upload controller code to ESP8266
	ESPAsyncWebServer	Asynchronous Web Server
	WiFi SoftAP	For ESP8266 to act as a Wi-Fi access point

Source Code

```
#include <Servo.h>
#include <ESP8266WiFi.h>
#include <ESPAsyncWebServer.h>

const char* ssid = "Team12";
const char* pw = "anuniyuv";

IPAddress local_ip(192,168,2,1);
IPAddress gateway(192,168,2,1);
IPAddress subnet(255,255,255,0);

AsyncWebServer server(80);

int light = D0;
int ac = D1;
int echo = D7;
int trigger = D8;
int temp = A0;
Servo door;

float timing, distance, value, room_temp;
int ac_temp = 24, intensity = 0, intensity100 = 0;

String message = "No one is at the door.";

bool light_on = LOW;
bool door_open = LOW;
bool ac_on = LOW;

bool light_click = LOW;
bool door_click = LOW;

void setup() {
  Serial.begin(9600);

  pinMode(light, OUTPUT);
```

```
  pinMode(ac, OUTPUT);
  pinMode(echo, INPUT);
  pinMode(trigger, OUTPUT);
  door.attach(D2);

  WiFi.softAP(ssid, pw);
  WiFi.softAPConfig(local_ip, gateway, subnet);
  delay(100);

  server.on("/", HTTP_GET, [](AsyncWebServerRequest *request) {
    request->send(200, "text/html", home(light_on, intensity100, message,
door_open, ac_on, room_temp, ac_temp));
  });

  server.on("/lightOn", HTTP_GET, [](AsyncWebServerRequest *request) {
    light_on = HIGH;
    intensity = 153;
    intensity100 = 60;
    request->send(200, "text/html", home(light_on, intensity100, message,
door_open, ac_on, room_temp, ac_temp));
  });

  server.on("/lightOff", HTTP_GET, [](AsyncWebServerRequest *request) {
    light_on = LOW;
    intensity = 0;
    intensity100 = 0;
    request->send(200, "text/html", home(light_on, intensity100, message,
door_open, ac_on, room_temp, ac_temp));
  });

  server.on("/adjustIntensityMinus", HTTP_GET, [](AsyncWebServerRequest
*request) {
    if (intensity > 0 && light_on)
      intensity -= 51;
    if (intensity == 0)
      light_on = LOW;
    intensity100 = (int)(intensity / 2.55);
    request->send(200, "text/html", home(light_on, intensity100, message,
door_open, ac_on, room_temp, ac_temp));
  });
```

```
  server.on("/adjustIntensityPlus", HTTP_GET, [](AsyncWebServerRequest
*request) {
    if (intensity < 255 && light_on)
      intensity += 51;
    intensity100 = (int)(intensity / 2.55);
    request->send(200, "text/html", home(light_on, intensity100, message,
door_open, ac_on, room_temp, ac_temp));
  });

  server.on("/doorClose", HTTP_GET, [](AsyncWebServerRequest *request) {
    door_open = LOW;
    request->send(200, "text/html", home(light_on, intensity100, message,
door_open, ac_on, room_temp, ac_temp));
  });

  server.on("/doorOpen", HTTP_GET, [](AsyncWebServerRequest *request) {
    light_on = HIGH;
    intensity = 153;
    intensity100 = 60;
    door_open = HIGH;
    request->send(200, "text/html", home(light_on, intensity100, message,
door_open, ac_on, room_temp, ac_temp));
  });

  server.on("/acOff", HTTP_GET, [](AsyncWebServerRequest *request) {
    ac_on = LOW;
    request->send(200, "text/html", home(light_on, intensity100, message,
door_open, ac_on, room_temp, ac_temp));
  });

  server.on("/acOn", HTTP_GET, [](AsyncWebServerRequest *request) {
    ac_on = HIGH;
    request->send(200, "text/html", home(light_on, intensity100, message,
door_open, ac_on, room_temp, ac_temp));
  });

  server.on("/adjustTempMinus", HTTP_GET, [](AsyncWebServerRequest
*request) {
    light_click = LOW;
```

```
    door_click = LOW;
    if (ac_temp > 16 && ac_on)
      ac_temp--;
    request->send(200, "text/html", home(light_on, intensity100, message,
door_open, ac_on, room_temp, ac_temp));
  });

  server.on("/adjustTempPlus", HTTP_GET, [](AsyncWebServerRequest
*request) {
    light_click = LOW;
    door_click = LOW;
    if (ac_temp < 30 && ac_on)
      ac_temp++;
    request->send(200, "text/html", home(light_on, intensity100, message,
door_open, ac_on, room_temp, ac_temp));
  });

  server.onNotFound([](AsyncWebServerRequest *request) {
    request->send(404, "text/plain", "Not found");
  });

  server.begin();
  Serial.println("HTTP server started.");
}

void loop() {
  digitalWrite(trigger, LOW);
  delay(2);
  digitalWrite(trigger, HIGH);
  delay(10);
  digitalWrite(trigger, LOW);

  timing = pulseIn(echo, HIGH);
  distance = (timing * 0.034) / 2;
  if (distance <= 10) {
    Serial.print("Distance: ");
    Serial.println(distance);
    message = "Motion detected. Open door?";
  } else {
    message = "No one is at the door.";
```

```
  }

  if (door_open && light_on) {
    door.write(270);
    delay(5000);
    analogWrite(light, intensity);
  } else {
    door.write(0);
    analogWrite(light, intensity);
  }

  if (light_on) {
    analogWrite(light, intensity);
  } else {
    analogWrite(light, intensity);
  }

  if (ac_on) {
    digitalWrite(ac, HIGH);
  } else {
    digitalWrite(ac, LOW);
  }
}

String home(bool light_on, int intensity, String message, bool door_open, bool
ac_on, float room_temp, int ac_temp) {
  String html;
  html.reserve(1500);
  html += "<!DOCTYPE html>";
  html += "<html>";
  html += "<head>";
  html += "<title>Residential Automation</title>";
  html += "<style>";
  html += ".header-container {";
  html += "    background-color: #2283d3;";
  html += "    color: white;";
  html += "    padding: 1rem;";
  html += "    display: flex;";
  html += "    justify-content: space-between;";
  html += "    align-items: center;";
```

```
html += "}";

html += ".header-container h1 {";
html += "    margin: 0;";
html += "    font-size: 2rem;";
html += "}";

html += ".main-container {";
html += "    display: flex;";
html += "    flex-direction: column;";
html += "    background-color: #ccebf1;";
html += "    border-radius: 10px;";
html += "    max-width: 100%;";
html += "    box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);";
html += "    margin: 1rem;";
html += "}";

html += ".form {";
html += "    display: flex;";
html += "    flex-direction: column;";
html += "    width: 100%;";
html += "}";

html += ".form label {";
html += "    font-weight: bold;";
html += "    margin-top: 1rem;";
html += "}";

html += ".form input {";
html += "    padding: 0.5rem;";
html += "    margin-top: 0.3rem;";
html += "    border: 1px solid #cccccc;";
html += "    border-radius: 5px;";
html += "    font-size: 1rem;";
html += "}";

html += ".form input:focus {";
html += "    outline: none;";
html += "    border-color: #2283d3;";
html += "    box-shadow: 0 0 5px #2283d3;";
```

```
html += "}";

html += ".btn {";
html += "    width: 100%;";
html += "    margin-top: 1rem;";
html += "    background-color: #2283d3;";
html += "    color: white;";
html += "    border: none;";
html += "    padding: 1rem;";
html += "    font-size: 1rem;";
html += "    cursor: pointer;";
html += "    border-radius: 10px;";
html += "}";

html += ".btn:hover {";
html += "    background-color: #2b92e6;";
html += "}";

html += "p {";
html += "    font-size: 1rem;";
html += "    font-weight: bold;";
html += "    color: #2283d3;";
html += "    margin-top: 1rem;";
html += "}";

html += ".slider {";
html += "    width: 100%;";
html += "    accent-color: #2283d3;";
html += "}";

html += "#intensityValue {";
html += "    font-weight: bold;";
html += "    color: #2283d3;";
html += "    margin-left: 0.5rem;";
html += "    font-size: 1rem;";
html += "}";

html += ".change-btn {";
html += "    padding: 0.5rem 1rem;";
html += "    font-size: 1rem;";
```

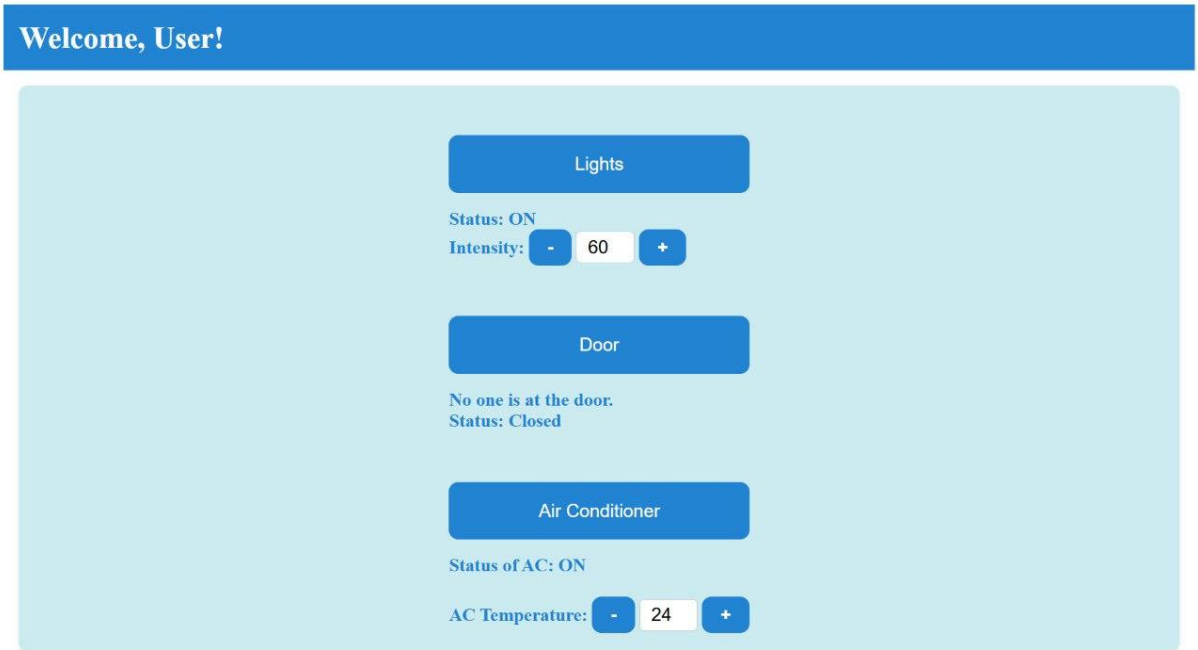
```
html += "    font-weight: bold;";
html += "    background-color: #2283d3;";
html += "    color: white;";
html += "    border: none;";
html += "    border-radius: 10px;";
html += "    cursor: pointer;";
html += "}";

html += ".change-btn:hover {";
html += "    background-color: #2b92e6;";
html += "}";

html += ".change-input {";
html += "    width: 50px;";
html += "    text-align: center;";
html += "    padding: 0.3rem;";
html += "    font-size: 1rem;";
html += "    border: 1px solid #ccc;";
html += "    border-radius: 5px;";
html += "}";

html += "@media (min-width: 768px) {";
html += "    .main-container {";
html += "        flex-direction: row;";
html += "        justify-content: center;";
html += "    }";

html += "    .btn {";
html += "        font-size: 1.2rem;";
html += "        padding: 1.2rem;";
html += "    }";

html += "    p {";
html += "        font-size: 1.2rem;";
html += "    }";

html += "    .change-btn {";
html += "        font-size: 1.2rem;";
html += "        padding: 0.5rem 1.2rem;";
html += "    }";
```

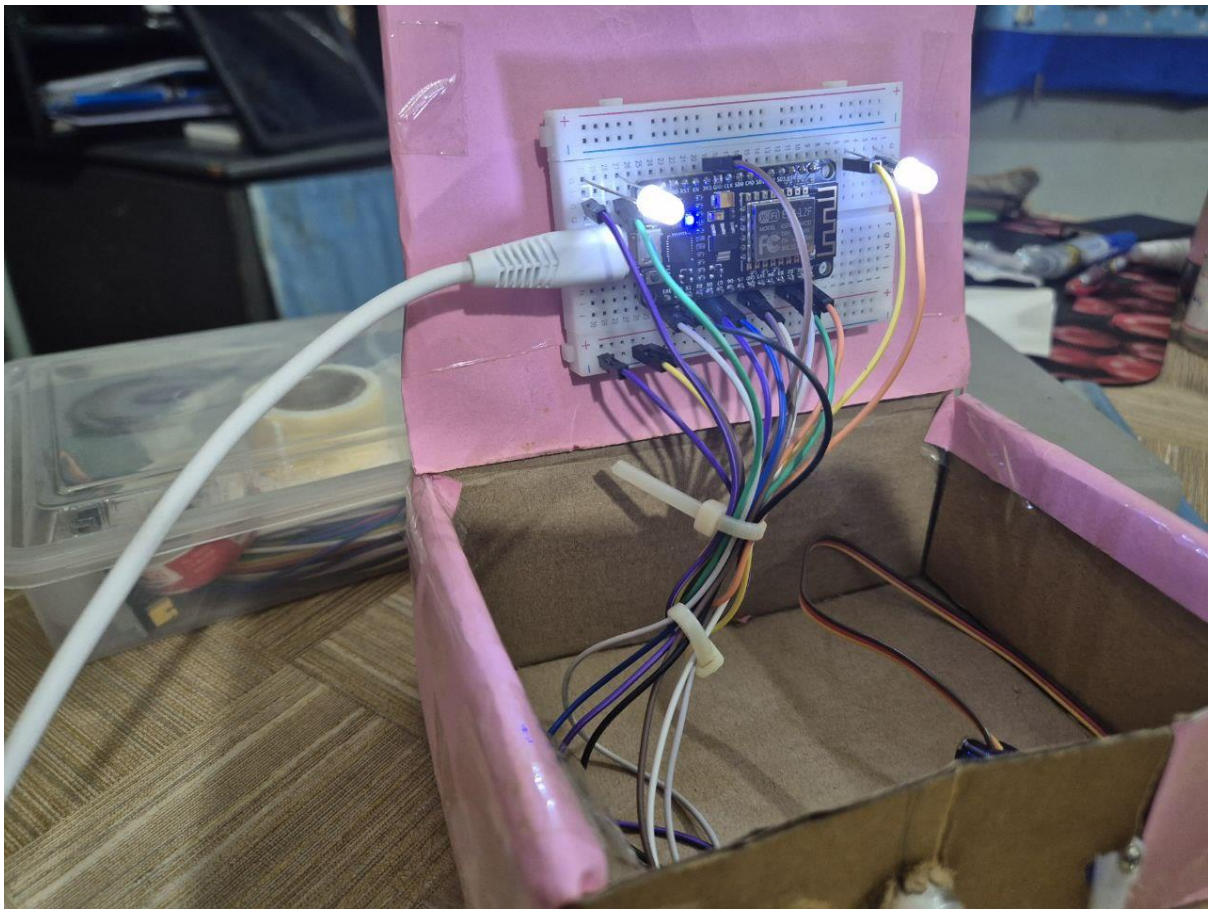
```
html += "    .change-input {";
html += "        font-size: 1.2rem;";
html += "    }";
html += "}";

html += "@media (max-width: 480px) {";
html += "    .header-container h1 {";
html += "        font-size: 1.5rem;";
html += "    }";

html += "    .btn {";
html += "        font-size: 0.9rem;";
html += "        padding: 0.8rem;";
html += "    }";

html += "    p {";
html += "        font-size: 0.9rem;";
html += "    }";

html += "    .change-btn {";
html += "        font-size: 0.9rem;";
html += "        padding: 0.3rem 0.8rem;";
html += "    }";

html += "    .change-input {";
html += "        font-size: 0.9rem;";
html += "    }";
html += "}";
html += "</style>";
html += "</head>";
html += "<body>";
html += "<header class='header-container'>";
html += "<h1>Welcome, User!</h1>";
html += "</header>";
html += "<main class='main-container'>";
html += "<form>";
html += "<br><br>";
if (light_on) {
```

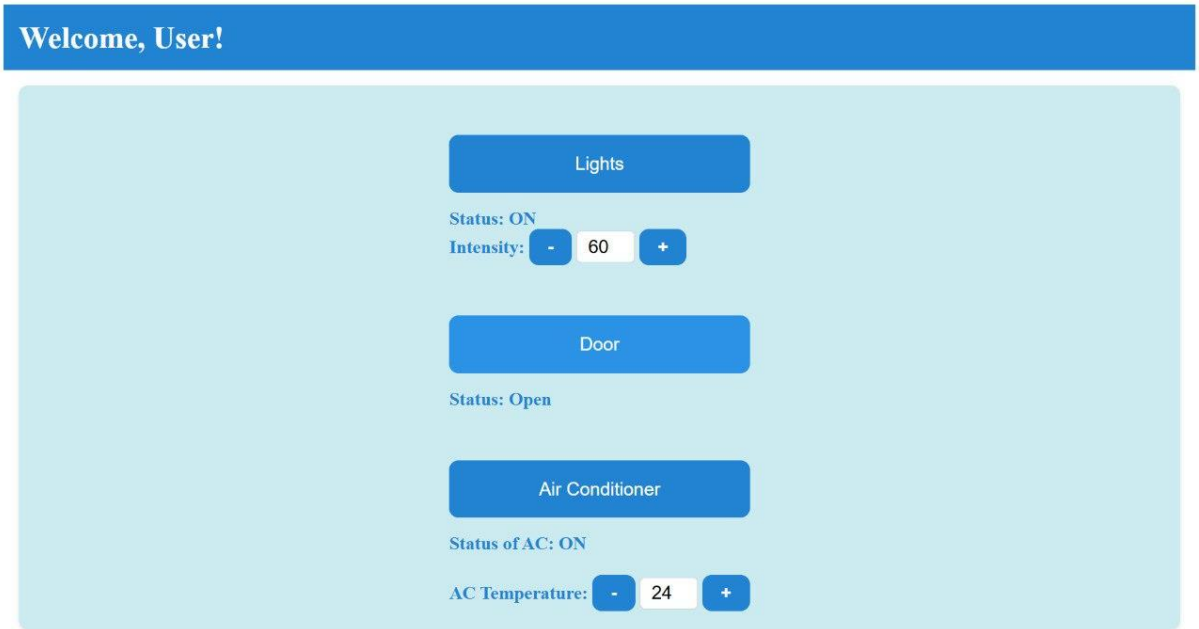
```
    html += "<input type='button' class='btn' value='Lights'
onclick=\"window.location.href='/lightOff'\">";
    html += "<p>Status: ON<br>";
  } else {
    html += "<input type='button' class='btn' value='Lights'
onclick=\"window.location.href='/lightOn'\">";
    html += "<p>Status: OFF<br>";
  }
  html += "Intensity: ";
  html += "<button type='button' class='change-btn'
onclick=\"window.location.href='/adjustIntensityMinus'\"'>-</button>";
  html += " <input type='number' class='change-input' value='";
  html += String(intensity);
  html += "' min='16' max='30' readonly> ";
  html += "<button type='button' class='change-btn'
onclick=\"window.location.href='/adjustIntensityPlus'\"'>+</button>";
  html += "</p><br>";
  if (door_open) {
    html += "<input type='button' class='btn' value='Door'
onclick=\"window.location.href='/doorClose'\">";
    html += "<p>Status: Open";
  } else {
    html += "<input type='button' class='btn' value='Door'
onclick=\"window.location.href='/doorOpen'\">";
    html += "<p>";
    html += message;
    html += "<br>Status: Closed";
  }
  html += "</p><br>";
  if (ac_on)
    html += "<input type='button' class='btn' value='Air Conditioner'
onclick=\"window.location.href='/acOff'\">";
  else
    html += "<input type='button' class='btn' value='Air Conditioner'
onclick=\"window.location.href='/acOn'\">";
  html += "<p>";
  html += "Room Temperature: ";
  html += String(room_temp);
  html += " &degC<br>";
  if (ac_on)
```

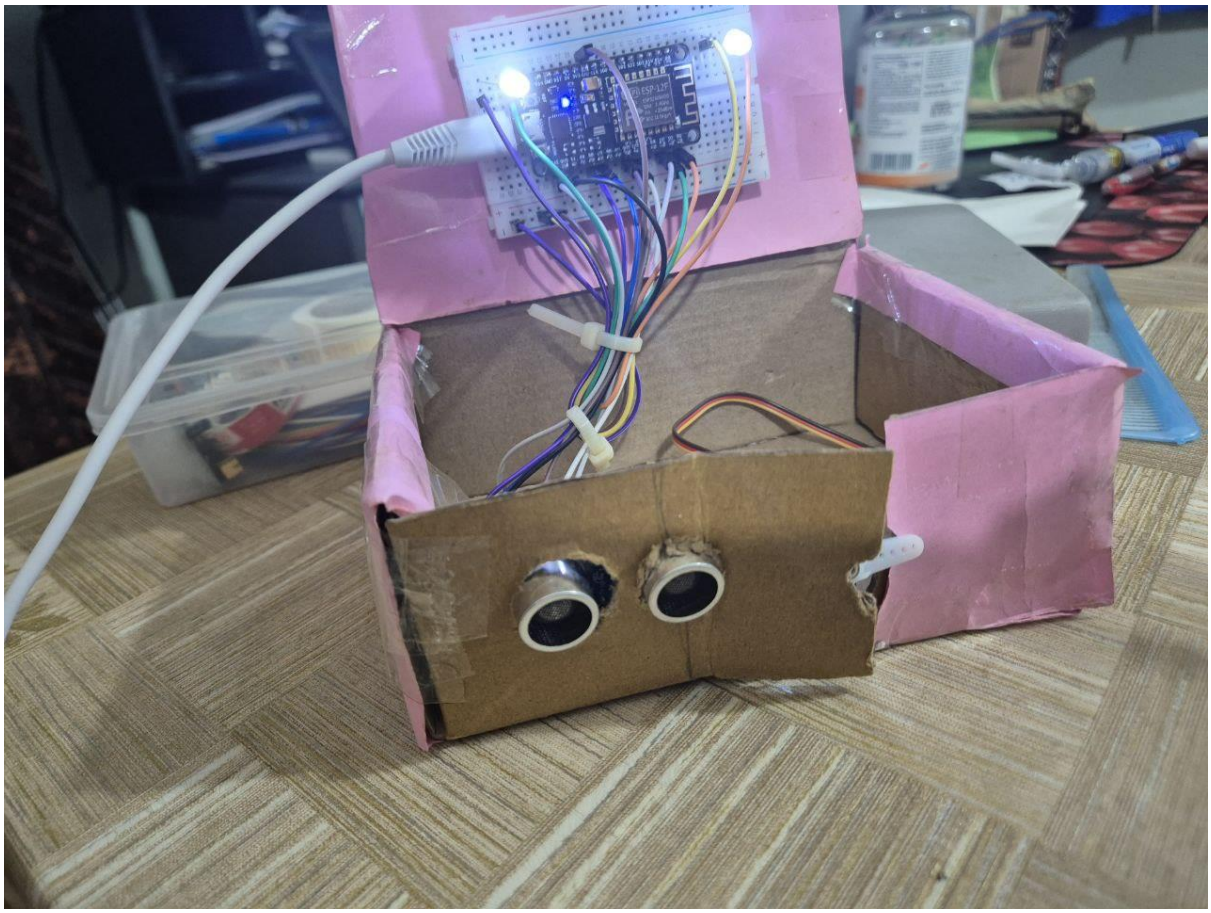
```
    html += "Status of AC: ON<br><br>";
  else
    html += "Status of AC: OFF<br><br>";
  html += "AC Temperature: ";
  html += "<button type='button' class='change-btn'
onclick=\"window.location.href='/adjustTempMinus'\"">-</button>";
  html += " <input type='number' class='change-input' value="";
  html += String(ac_temp);
  html += "' min='16' max='30' readonly> ";
  html += "<button type='button' class='change-btn'
onclick=\"window.location.href='/adjustTempPlus'\"">+</button>";
  html += "</p>";
  html += "</form>";
  html += "</main>";
  html += "</body>";
  html += "<br><br><br>";
  html += "</html>";
  return html;
}
```

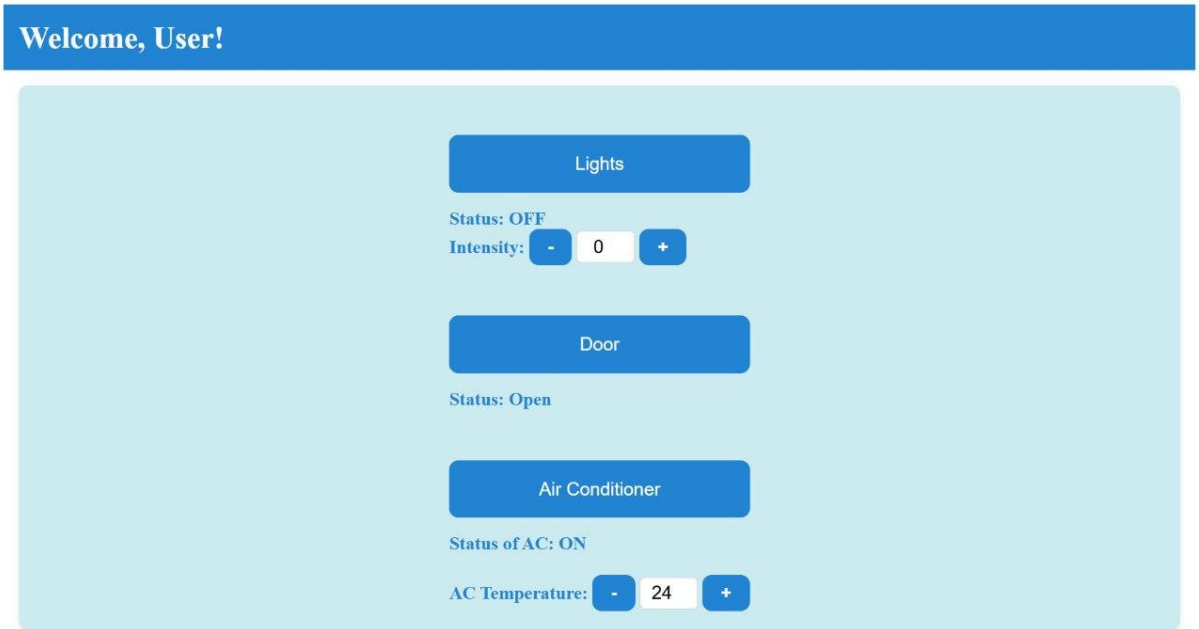
Output

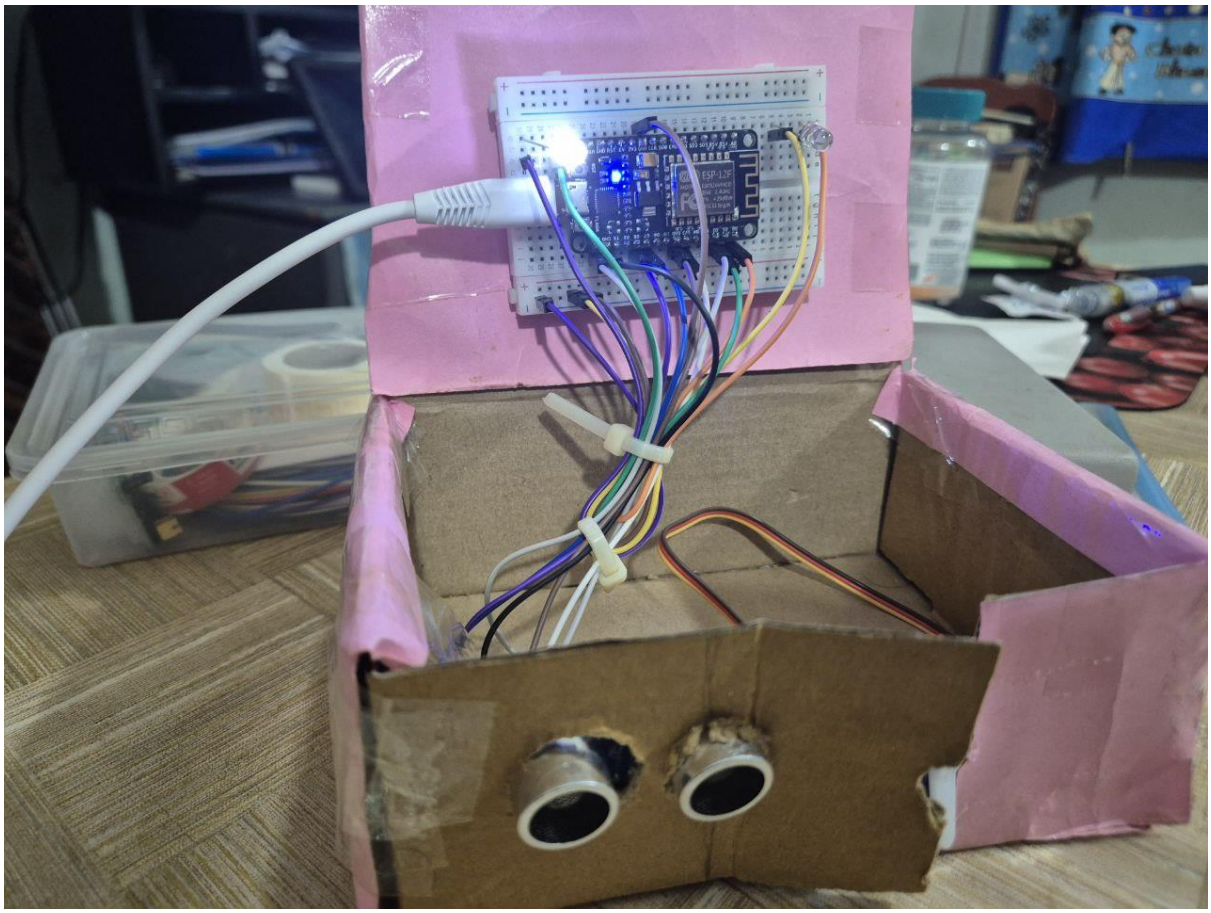

Welcome, User!
Lights
Status: ON
Intensity:
-
60
+
Door
No one is at the door.
Status: Closed
Air Conditioner
Status of AC: ON
AC Temperature:
-
24
+




Welcome, User!
Lights
Status: ON
Intensity:
-
60
+
Door
Status: Open
Air Conditioner
Status of AC: ON
AC Temperature:
-
24
+




Welcome, User!
Lights
Status: OFF
Intensity:
-
0
+
Door
Status: Open
Air Conditioner
Status of AC: ON
AC Temperature:
-
24
+



Potential Improvements

The most obvious improvement that can be made to CIRA is to add more devices, which of course, given the limited structural nature of the prototype setup, could be too complex to implement. However, on an actual home that is more than spacious enough to accommodate all the necessary devices, providing smart, automated functionality with effective wiring management should be not just possible, but also easy.

The conversational interface is limited to fixed commands whose wordings can't be changed. This obviously provides a barrier to us human users, who would obviously want more natural communication. Natural language processing techniques and AI-enabled conversations that understand not just simple commands but full-blown statements from human users who could be very vague or very descriptive are features that would bring immense quality of life improvements and further convince old-school regular home diehards to switch to what is most certainly the better solution, at least in an ideal world.

There are many more improvements that can be made. As time progresses and with better adoption, more flaws or limitations to this implementation could be found and discovered, which is not a bad thing – it is only by understanding these flaws that the near-perfect version of the desired dream can be achieved.

It is important to close out all the possible loopholes if and when they arise. More real world testers, more errors, and more fixes pave the way not just to eventual stability but also the ideal smart home environment envisioned through CIRA.

Conclusion

The Conversation Interface for Residential Automation, or CIRA, is an effective and efficient smart home automation tool that provides simple tap and play functionality to control and manipulate the various smart devices that are present at one’s home.

It integrates different types of devices with divergent functionalities into a seamless web and mobile interface that is fast and responsive.

The developed prototype visualises a formidable smart home system that has the capability to enhance the lives of people, automating different parts of their homes in a meaningful, sensible, and understandable manner.

It aims to avoid the roadblocks and flaws that current implementations have, providing not an alternative, but a reimagination and improvement on what already exists. It connects very well with existing implementations, making it not too hard at all to successfully bring about on a large scale.

CIRA is not perfect, but it is better, and can be even more so with the proper usage and feedback cycle that we expect to receive for any good project.